Parts List – Class D Audio Amplifier

Components required, with quantities. Please confirm availability and price.

#	Component	Qty	Notes
Power			
1	12 V 1 A DC adapter	1	Already owned
2	DC barrel jack, PCB or panel mount, matching the adapter plug	1	
3	1N5819 Schottky diode	1	Reverse-polarity protection, in series with supply
4	SPST slide switch	1	Optional
5	LED + 1 kΩ resistor	1 + 1	Power indicator
Audio input			
6	Type-C to 3.5 mm audio adapter	1	External, like the phone
7	3.5 mm stereo jack, PCB mount	1	
8	10 kΩ log (audio taper) potentiometer	1	Volume
9	1 µF capacitor	1	Input coupling
ICs			
10	TL074 (quad op-amp, DIP-14)	1	Preamp, triangle oscillator, mid-rail buffer
11	LM393 (comparator, DIP-8)	1	Audio vs triangle gives PWM
12	IR2104 (half-bridge driver, DIP-8)	1	
13	IC sockets: 14-pin ×1, 8-pin ×2	3	
Resistors, trimmer, small capacitors			
14	10 kΩ resistor, ¼ W	8	
15	22 kΩ resistor, ¼ W	1	
16	100 kΩ trimmer	1	Preamp gain
17	10 Ω resistor, ¼ W	2	MOSFET gate resistors
18	1 nF film or NP0 capacitor	1	Sets ~50 kHz triangle
19	10 µF, 25 V electrolytic capacitor	1	Mid-rail filter
20	0.1 µF ceramic capacitor	4	Decoupling
Power stage			
21	IRFZ44N or IRLZ44N MOSFET, TO-220	2	
22	UF4007 or FR107 fast diode	1	Bootstrap
23	1 µF, 25 V capacitor	1	Bootstrap
24	470 µF, 25 V electrolytic capacitor	1	Bulk supply at the MOSFETs
25	Small clip-on heatsink, TO-220	2	Optional
Output filter and speaker			
26	47 µH power inductor, 3 A or more	1	
27	1 µF, 63 V polyester film capacitor	1	
28	1000 µF, 25 V electrolytic capacitor	1	DC-blocking output cap
29	10 Ω 1 W resistor + 100 nF capacitor	1 + 1	Snubber, optional
30	4 Ω 5 W speaker	1	Already owned
31	2-pin screw terminal	1	